Optical Properties of a Transition-Metal Dichalcogenide ZnO Nanowire Field-Effect Transistor

Y. Bulsara[1,], M. Tomoscheit[1], O. Ghaebi[1], M. Willems[1], E. Eobaldt[1], C. Ronning[1], and G. Soavi[1,2]*

[1]*Institute of Solid State Physics, Friedrich Schiller University Jena, Jena, Germany*

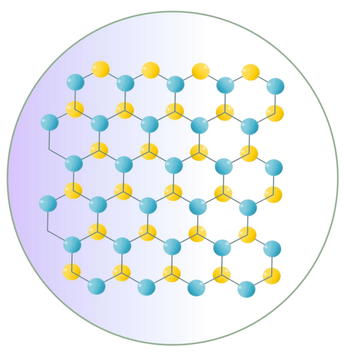
[2]*Abbe Center of Photonics, Friedrich Schiller University Jena, Jena, Germany*

**Email: yashvi.bulsara@uni-jena.de*

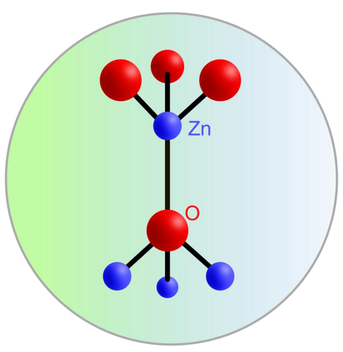
Motivation

Nanoscale materials offer powerful opportunities for advancing photonic and optoelectronic technologies, where compact coherent light sources are essential for integrated photonic circuits. Hybrid 1D-2D semiconductor heterostructures provide a promising route toward electrically switchable and tunable nanolasers.

- ZnO nanowires: Natural nanoscale lasers (optical gain, amplification and waveguiding)
- ML-WSe_2: Strong light-matter interaction, electrostatic tuning [1][2]

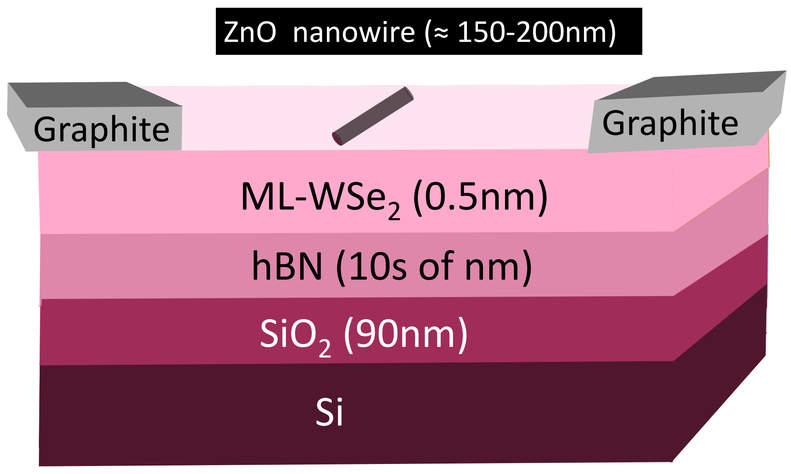


MONOLAYER WSe_2

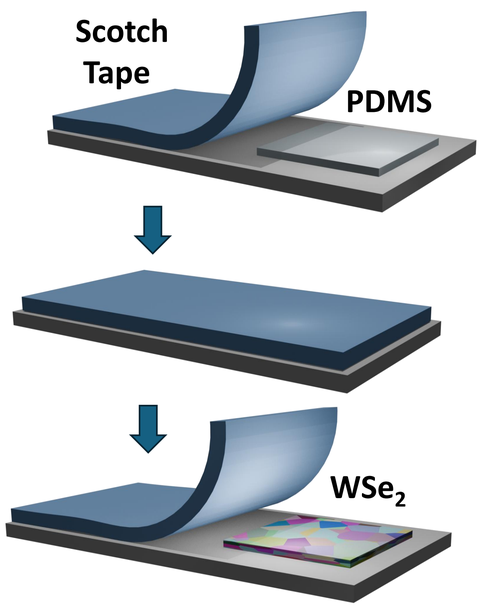


ZnO NANOWIRE

- Hybrid ZnO-WSe_2: Electrostatic modulation, control the lasing [3]

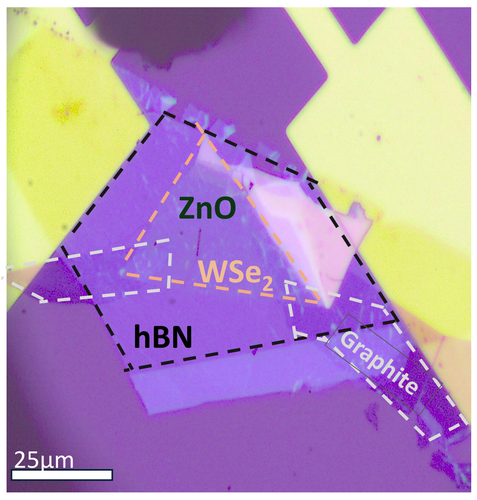


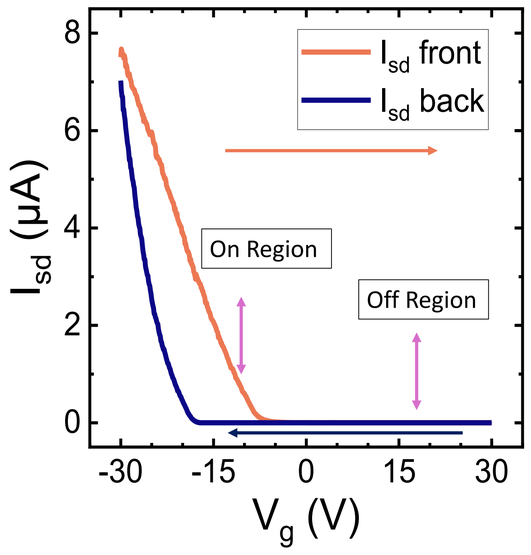
Sample Fabrication



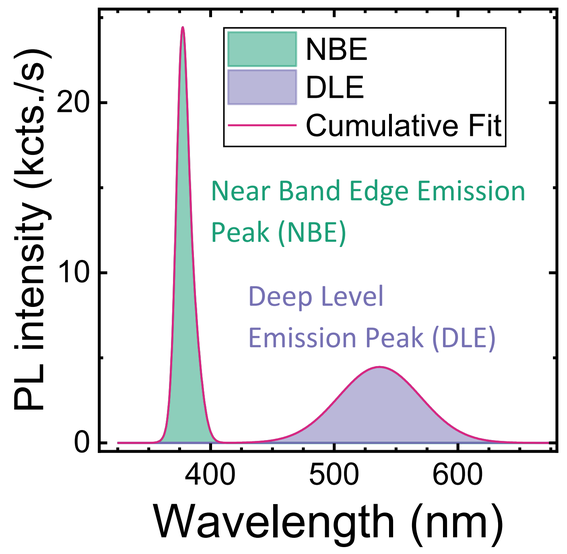
- **Transfer of ML-WSe_2 on PDMS**
- **Heating for T=50° C for 5 minutes**
- **Results: Large ML-WSe_2 flakes**

Device Architecture and Characterization

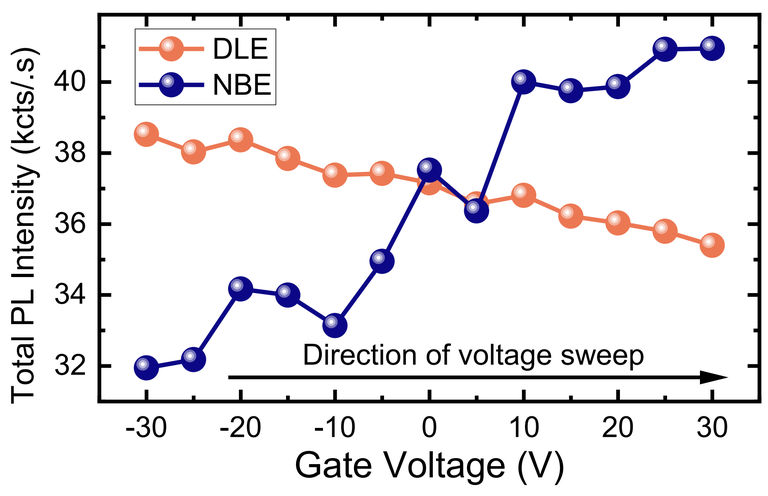


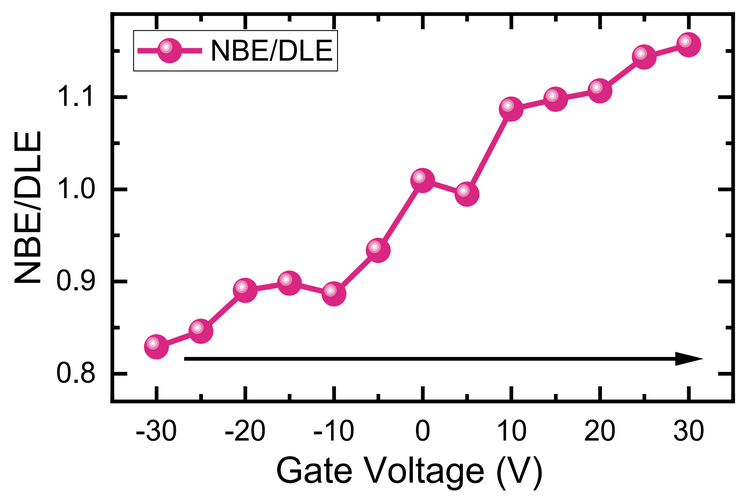


Photoluminescence



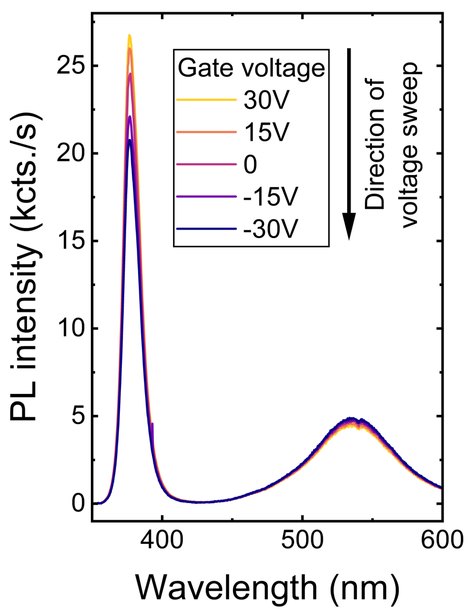
Hybrid-Photoluminescence



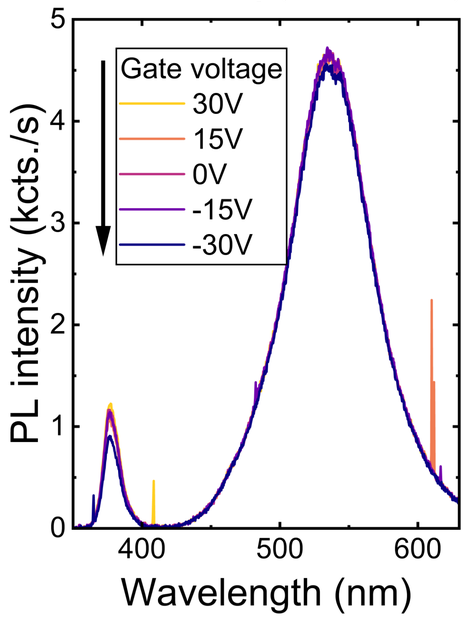


Device Stability

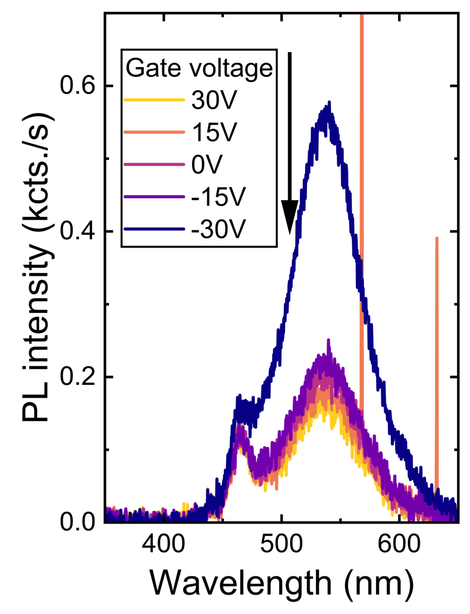
Forward Sweep (-30 V to +30V)



Backward Sweep (+30 V to -30V)



Gate Dependence after Vacuum Treatment



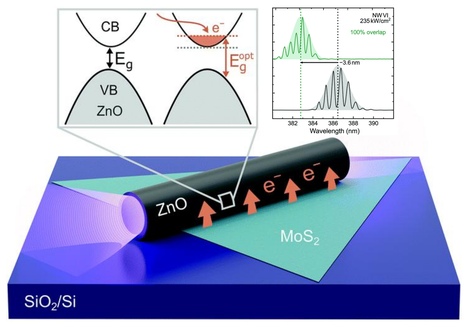
Conclusion and Outlook

Conclusion:

- Successful fabrication of a ZnO-WSe_2 hybrid field-effect transistor
- Gate voltage modulates nanowire PL, indicating ZnO-WSe_2 interaction
- Repeated experiments show limited device stability

Outlook:

- Improve device stability by studying possible reasons (trapping, UV bleaching [4])
- High-vacuum measurements to reduce hysteresis
- Band alignment and electrically tunable lasing



Adapted from [3] with permission

References

[1] Roadmap for Photonics with 2D Materials, *ACS Photonics*, 12, 3961–4095 (2025).
[2] O. Ghaebi et al., *ACS Nano*, 19, 19027–19034 (2025).
[3] E. Eobaldt et al., *Nanoscale*, 14, 6822–6829 (2022).
[4] A. George et al., *npj 2D Mater. Appl.*, 5, 15 (2021).

NOA NONLINEAR OPTICS DOWN TO ATOMIC SCALES

Faculty of Physics and Astronomy
FRIEDRICH-SCHILLER-UNIVERSITÄT JENA